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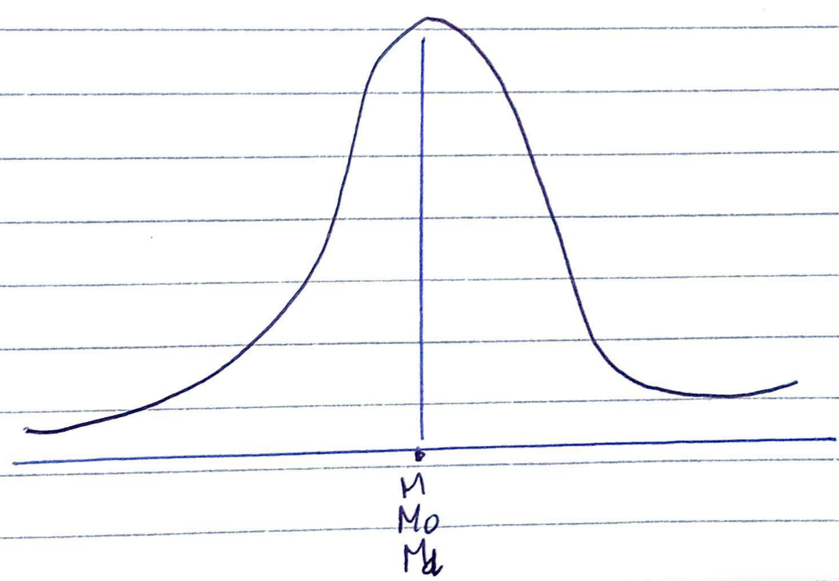
Skewness

For symmetrical distribution, the frequencies are symmetrically distributed about the ~~man~~ mean i.e., variates equidistant from the mean have equal frequencies. Also, the mean, mode and median coincide and median lies halfway between the two quartiles. Thus, $M = M_o = M_d$ and ~~$Q_3 \neq$~~

$$Q_3 - M = M - Q_1.$$

Note: Symmetrical Distribution:

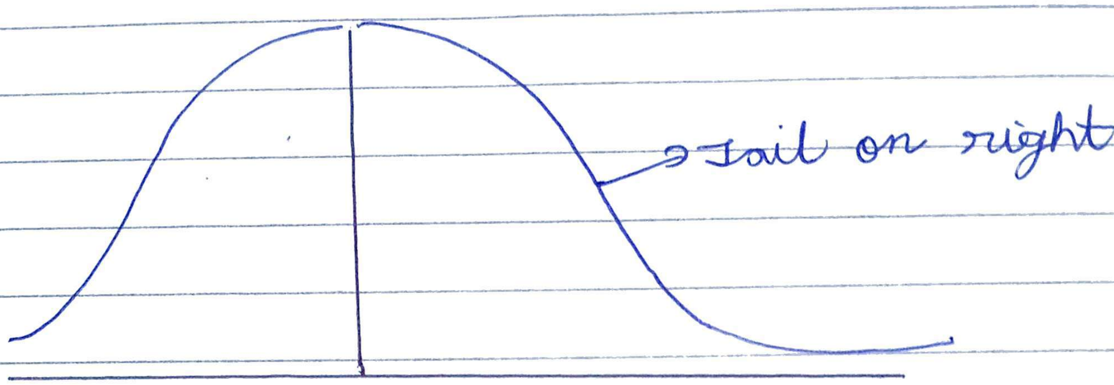


Meaning of skewness: If the curve of the distribution is not symmetrical, it may admit of tail on either side of distribution. Skewness means lack of symmetry.

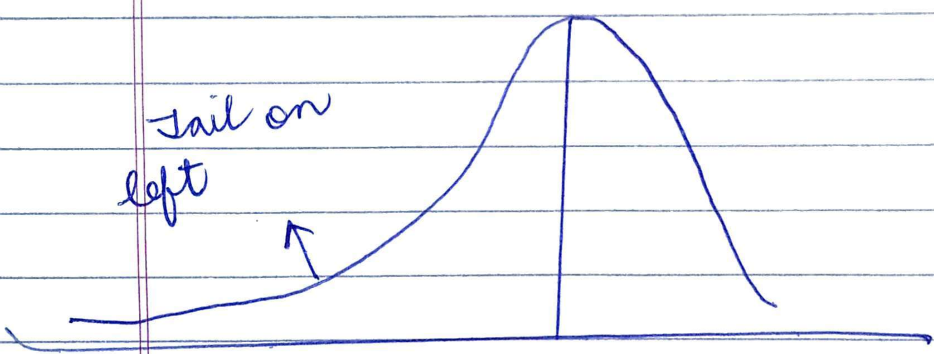
Note: Skewness indicates whether the curve is turned more to one side than to other i.e., whether the curve has a longer tail on one side.

Note: Skewness can be positive as well as negative.

Positive Skew Distribution:



Negative Skew Distribution:



Test of skewness:

(i) If AM = Mode = Median, then, there is no skewness.

(ii) If AM < the value of mode, the tail would be on a left.

(iii) If AM > the value of mode, the tail would be on right.

(iv) If the sum of frequencies of values < mode is equal to the sum of frequencies of values > mode, then there would be no skewness.

(v) If the quartiles are equidistant from median, then there could be no skewness

Method of measuring skewness

1 Relative measures of skewness are called the coefficient of skewness. They are independent of the units of measurement, and as such, they are pure numbers.

2. The following are the methods of measuring skewness:

i) Karl Pearson Method
ii) Bowley's Method
iii) Kelly's Method
iv) Method of moments

i) Karl Pearson Method: This method is based on the fact that in a symmetrical distribution, the value of A.M. is equal to that of a mode. As we have already noted that, the distribution is positively skewed of arithmetic mean > mode and negatively skewed if A.M. < mode.

The Karl-Pearson coefficient of skewness is given by $\frac{A.M - Mode}{S.D.}$

Note: If mode is ill-defined in some frequencies distribution, then the value of empirical mode is used in the formula

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Empirical Mode = 3 Median − 2 Arithmetic Mean.

Coefficient of skewness : $\frac{A.M. - Mode}{S.D.}$

$= \frac{AM - (3 Median - 2A.M)}{S.D}$

$= \frac{A.M - 3 Median + 2A.M}{S.D}$

$= \frac{3 (A.M - Median)}{S.D}$.

this is the formula of Karl Pearson Coefficient.

Assignment : i) Write the three real world problems and their solutions by using skewness.
ii) Write ~~Sk~~ and express application of skewness in AI Era.
iii) Write the application of skewness in field of Health Sciences.

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Ex-1. Karl Pearson's coefficient of skewness of a distribution is 0·32, its standard deviation is 6·5 and mean is 29·6. Find the mode of the distribution.

⟹ We have, $Sk_p = 0.32$, $S.D. = 6.5$, $\bar{x} = 29.6$

Now, $Sk_p = \frac{\bar{x} - Mode}{S.D.}$

$0.32 = \frac{29.6 - Mode}{6.5}$

$2.08 = 29.6 - Mode$

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Mode $= 29.6 - 2.08 = 27.52$

Ex-2: For a moderately skewed data, the arithmetic mean is 100, the variance is 35 and Karl Pearson's coefficient of skewness is 0.2. Find its mode and median.

⇒ Given, $\bar{x} = 100$, Variance $= 35$, $K_p = 0.2$

Now,
$$K_p = \frac{\bar{x} - \text{Mode}}{\sigma}$$

$$0.2 = \frac{100 - \text{Mode}}{\sqrt{35}}$$

$$0.2 \times 5.92 = 100 - \text{Mode}$$

$$1.184 = 100 - \text{Mode}$$

$$\text{Mode} = 98.816.$$

Also,
$$\text{Mode} = 3\,\text{Median} - 2AM$$

$$98.816 = 3\,\text{Median} - 2(100)$$

$$\text{Median} = \frac{298.816}{3} = 99.61.$$

Ex-3. In a certain distribution, the following results were obtained: AM $= 45$, Median $= 48$, Coefficient of skewness $= -0.4$. The person who gave you this data, failed to give the value of S.D. You are required to estimate it with the help of available data.

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$\Rightarrow$ Given, coeff. of skewness, $K_p = -0.4$, A.M $= 45$, Median $= 48$

Now, $$K_p = \frac{3(A.M - Median)}{S.D.}$$

$$\frac{-4}{10} = \frac{3(45-48)}{S.D}$$

$$\frac{-4}{10} = \frac{3(-3)}{S.D} \Rightarrow S.D = \frac{9 \times 10}{4}$$

$$S.D = \frac{90}{4} = 22.5$$

Ex-4 The sum of 20 observations is 300 and sum of their squares is 5000. The median is 15. Find the Karl Pearson's coefficient of skeweness.

$\Rightarrow$ Given, $n = 20$, $\Sigma x = 300$, $\Sigma x^2 = 5000$, Median $= 15$.

Now, $$\bar{x} = \frac{\Sigma x}{n} = \frac{300}{20} = 15$$

$$\sigma^2 = \frac{\Sigma x^2}{n} - \bar{x}^2 = \frac{500\cancel{0}}{2\cancel{0}} - (15)^2$$

$$\sigma^2 = 250 - 225$$

$$\sigma^2 = 25$$

$$\therefore \boxed{\sigma = 5}$$

Now, Karl Pearson coeff. of skewness

$$Kp = \frac{3(\bar{x} - \text{Median})}{S.D.} = \frac{3(15-15)}{5} = 0$$

Assignment: Find the coefficient of skewness by Karl Pearson's method for the following data:

Value	6	12	18	24	30	36	42
Frequency	4	7	9	18	15	10	3

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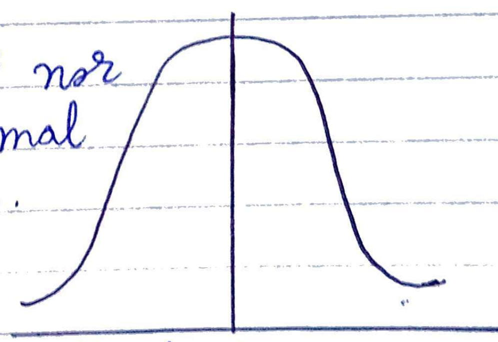
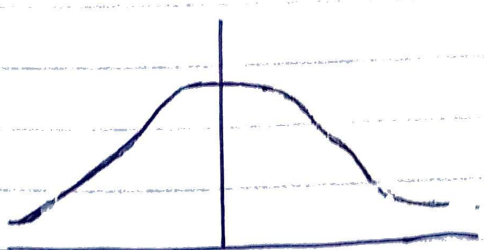
Kurtosis: The relative flatness of the top is called kurtosis and is measured by β_2.

① Curves which are neither flat nor sharply peaked are called normal curves or mesokurtic curves.



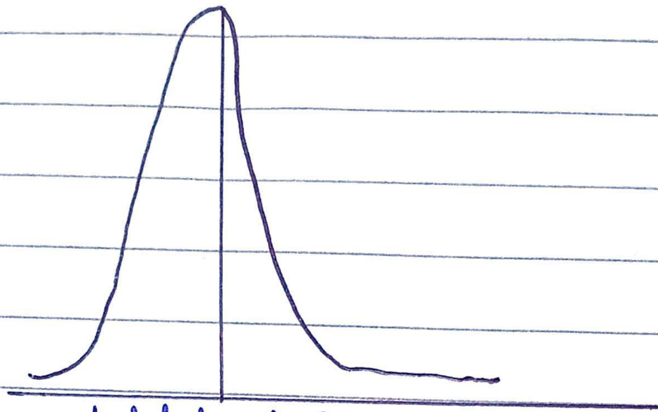
Mesokurtic
$\beta_2 = 3$
$\gamma_2 = 0$

② 

Platykurtic
$\beta_2 < 3$
$\gamma_2 < 0$

Curves which are flatter than the normal curve are called platykurtic curves.

3) Curves which are more sharply peaked than the normal curve are called leptokurtic curves.



Leptokurtic
$\beta_2 > 3$
$\gamma_2 > 0$

Measure of Kurtosis

The measure of kurtosis is denoted by β_2 and is defined as

$$\boxed{\beta_2 = \frac{\mu_4}{\mu_2^2}}$$

where μ_2 and μ_4 are respectively the second and fourth moments about the mean of the distribution.

Note

i) If $\beta_2 > 3$, the distribution is leptokurtic.
ii) If $\beta_2 = 3$, the distribution is mesokurtic.
iii) If $\beta_2 < 3$, the distribution is platykurtic.

The kurtosis of a distribution is also measured by using Greek letter 'γ_2' which is defined

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as $\gamma_2 = \beta_2 - 3$.

Note:

i) If $\gamma_2 > 0$, the distribution is leptokurtic.
ii) If $\gamma_2 = 0$, the distribution is mesokurtic.
iii) If $\gamma_2 < 0$, the distribution is platykurtic.

Methods of calculation of Kurtosis

I. If the value of μ_2 and μ_4 are given, then find β_2 by using the formula: $\beta_2 = \frac{\mu_4}{\mu_2^2}$

II. If raw moments μ_1', μ_2', μ_3' and μ_4' are given, then calculate:

$$\mu_2 = \mu_2' - \mu_1'^2 \quad \text{and} \quad \mu_4 = \mu_4' - 4\mu_3'\mu_1' + 6\mu_2'\mu_1'^2 - 3\mu_1'^4.$$

Ex-1) The first four moments about mean of a frequency distribution are 0, 100, −7 and 35000. Discuss the kurtosis of the distribution.

⇒ Given, $\mu_1 = 0$, $\mu_2 = 100$, $\mu_3 = -7$, $\mu_4 = 35000$

Now, $\beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{35000}{(100)^2} = 3.5$

∵ $3.5 > 3$

∴ The distribution is leptokurtic.

Ex-2: The first four moments of a distribution about the value '4' of the variable are −1.5, 17, −30 and 108. State whether the distribution is leptokurtic or platykurtic.

⇒ Given, $\mu_1' = -1.5$, $\mu_2' = 17$, $\mu_3' = -30$, $\mu_4' = 108$

Moments about mean :

$$\mu_2 = \mu_2' - \mu_1'^2 = 17 - (-1.5)^2$$
$$= 17 - 2.25 = 14.75$$

and $$\mu_4 = \mu_4' - 4\mu_3'\mu_1' + 6\mu_2'\mu_1'^2 - 3\mu_1'^4$$

$$\mu_4 = 108 - 4\times(-30)\times(-1.5) + 6\times(17)\times(2.25) - 3\times(-1.5)^4$$
$$= 108 - 180 + 229.5 - 15.1875$$
$$= 142.3125$$

Kurtosis : $\beta_2 = \dfrac{\mu_4}{\mu_2^2} = \dfrac{142.3125}{(14.75)^2} = 0.6541$

Since, $\beta_2 < 3$, the distribution is platykurtic.

Q. The standard deviation of a symmetric distribution is 5. What must be the value of the fourth moment about the mean in order that the distribution be
(i) leptokurtic (ii) mesokurtic (iii) platykurtic?

$\Rightarrow$ Given, standard deviation $\sigma = 5$.

$\therefore$ Variance $\sigma^2 = 25 \Rightarrow \mu_2 = 25$

Now, $\beta_2 = \dfrac{\mu_4}{\mu_2^2} = \dfrac{\mu_4}{25^2} = \dfrac{\mu_4}{625}$

Thus, the distribution will be

(i) Leptokurtic if $\beta_2 > 3 \Rightarrow \dfrac{\mu_4}{625} > 3 \Rightarrow \mu_4 > 1875$

(ii) Mesokurtic if $\beta_2 = 3 \Rightarrow \dfrac{\mu_4}{625} = 3 \Rightarrow \mu_4 = 1875$

(iii) Platykurtic if $\beta_2 < 3 \Rightarrow \dfrac{\mu_4}{625} < 3 \Rightarrow \mu_3 < 1875$.